

Optic neuropathy: tempo and context shape the differential

Acuity, colour, pupils, fields and disc appearance should be interpreted together.

Important optic nerve patterns

Pattern	Typical clues
Optic neuritis	Subacute monocular loss; pain on movement; marked colour impairment.
NAION	Sudden painless waking loss; vascular risks; hyperaemic swollen disc.
AAION	Profound loss; GCA symptoms; often chalky swollen disc.
Toxic or hereditary	Bilateral central loss; medication or maternal family context.

Clinical meaning

Interpret acuity, colour, RAPD, fields and disc together. A normal disc does not exclude retrobulbar or toxic disease, and symmetric bilateral loss can reduce RAPD. MRI assesses demyelination and compression. Suspected AAION needs urgent systemic steroids and GCA assessment to protect the fellow eye; suspected ethambutol toxicity requires prompt cessation.

Swollen discs are a finding before they are a diagnosis

Bilateral pressure-related swelling and unilateral inflammatory swelling need different explanations.



Neuroretinitis: the macular star is a radial pattern of hard exudates that may appear after initial disc swelling.

Clinical meaning

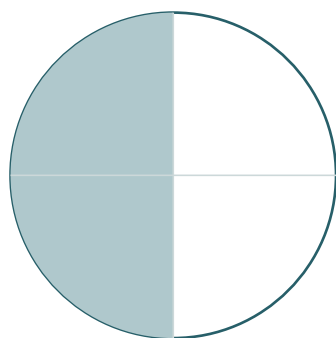
Papilloedema means swelling from raised ICP, often with preserved early acuity. Check BP and image the brain before appropriate LP to exclude a mass. Headache, tinnitus and transient obscurations support raised ICP. Sixth nerve palsy can be false localising. Neuroretinitis may show disc swelling before its macular star appears.

Localise visual space and identify the abnormal pupil

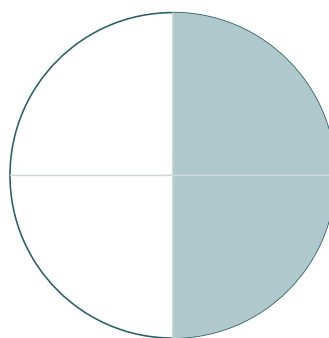
Bitemporal loss points to the chiasm; pupil asymmetry follows a separate efferent pathway.

Bitemporal visual field loss

Patient view; shaded regions represent impaired vision



Left eye



Right eye

Original schematic: both temporal fields are affected, consistent with chiasmal disease.

Clinical meaning

Homonymous loss affects the same half of visual space in both eyes: a right retrochiasmal lesion affects left space. Horner instead links same-side miosis and mild ptosis, with anisocoria worse in the dark. Adie has a large tonic pupil and light-near dissociation. Painful acute Horner or pupil-involving third nerve palsy needs urgent vascular assessment.

Diplopia can be paralytic, fatigable or restrictive

Orbital signs and optic nerve function determine whether the presentation is sight-threatening.

Three ways movements become abnormal

Nerve palsy

A stable weak movement can fit III, IV or VI distribution.

Ocular MG

Fluctuating painless diplopia and ptosis, with fatigue and pupil sparing.

Restriction

Fibrosis or orbital inflammation mechanically limits passive movement.

Clinical meaning

Assess acuity, colour and pupils beyond the diplopia. Orbital cellulitis needs urgent admission, IV antibiotics and CT assessment for abscess. New optic neuropathy in TED needs urgent treatment; exposure needs corneal protection. Ice-pack improvement supports MG. Ask about bulbar and respiratory weakness, and investigate associated thymic disease when appropriate.